STACK-IN-SETOIDS PROPERTY IS INVARIANT UNDER EQUIVALENCE OF CATEGORIES OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. The cited lemma-stack-in-setoids-characterize carries an 'Omitted' proof in the source; it is used here as a declared reference only.

Allowed standard prerequisites (editorial).

- Setoid fibres characterization. categories-lemma-setoid-fibres (cross-chapter, named). Characterizes fibred-in-setoids as being equivalent over the base to a category fibred in sets, so the property transfers along the given equivalence.
- Stack-in-setoids characterization. lemma-stack-in-setoids-characterize (same chapter, named). Reduces the stack-in-setoids condition to the associated category fibred in sets being a stack in sets.

Source reference. Original reference 0045: focused-seeds/geometry/S9-stk-stack-in-setoids-equivalent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0432: sources/stacks-stacks/pinned-original-source.tex, source lines 1331–1331. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{S}_1$, $\mathcal{S}_2$ be categories over $\mathcal{C}$. Suppose that $\mathcal{S}_1$ and $\mathcal{S}_2$ are equivalent as categories over $\mathcal{C}$. Then $\mathcal{S}_1$ is a stack in setoids over $\mathcal{C}$ if and only if $\mathcal{S}_2$ is a stack in setoids over $\mathcal{C}$.*

Proof. By Categories, Lemma 0045 we see that a category $\mathcal{S}$ over $\mathcal{C}$ is fibred in setoids over $\mathcal{C}$ if and only if it is equivalent over $\mathcal{C}$ to a category fibred in sets. Hence we see that $\mathcal{S}_1$ is fibred in setoids over $\mathcal{C}$ if and only if $\mathcal{S}_2$ is fibred in setoids over $\mathcal{C}$. Hence now the lemma follows from Lemma 0432. □

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